

CH1101: Quantum Chemistry and Spectroscopy

Tutorial Sheet 02

Module: Quantum Chemistry

- Q1. A helium-neon laser usually used in supermarket scanners emits light of wavelength 633 nm. Calculate the frequency of this light. What is the energy of a photon generated by this laser?
- Q2. When lithium is irradiated with light, the kinetic energy of the ejected electrons is given as: 2.935×10^{-19} J for $\lambda = 300.0$ nm, and 1.280×10^{-19} J for $\lambda = 400.0$ nm. Calculate the (a) Planck constant, (b) work function, and (b) threshold frequency of lithium from these data.
- Q3. Calculate the de Broglie wavelength for a baseball (0.14 kg) traveling at 144 km/h. Now find out the de Broglie wavelength of an electron traveling at 1.0 % of the speed of light. Interpret your results on these two moving objects.
[Note: Mass of an electron = 9.109×10^{-31} kg]